

LeafOS llama.cpp Vulkan Provider Spine

Runtime Boundary and Operator Checks

LeafOS

Purpose

This document records the intended local inference boundary for LeafOS: `llama.cpp` is the model provider, and Vulkan is the preferred local GPU backend. The provider is a long-lived `llama-server` process. It is not the node transport and it is not the filesystem authority.

Control Plane

```
Human task
-> task file
-> CPU graph and policy checks
-> provider call to llama.cpp HTTP endpoint
-> model proposal
-> parser
-> patch gate / verification gate
-> accepted artifact or rejection
-> journal / report
```

Provider Startup

On Windows, use the PowerShell provider launcher:

```
cd C:\Users\liamm\OneDrive\Documents\LeafOS\PowerShell-Version
.\serve-llamacpp.ps1 --llama-dir C:\llama-vulkan --backend vulkan --model fable
```

Use `-check-only` before a real run:

```
.\serve-llamacpp.ps1 --llama-dir C:\llama-vulkan --check-only --backend vulkan
```

Provider Environment

Set these in the shell that will run LeafOS commands:

```
$env:LEAF_PROVIDER_MODE = 'llamacpp'
$env:LEAF_BRAIN_PROVIDER = 'llamacpp'
$env:LEAF_CODER_PROVIDER = 'llamacpp'
$env:LEAF_LLAMACPP_URL = 'http://127.0.0.1:8080'
```

Diagnostics

Check	Meaning
<code>vulkaninfo -summary</code>	Confirms the OS Vulkan stack sees a GPU.
<code>llama-server -list-devices</code>	Confirms the selected llama.cpp build exposes accelerator devices.
<code>leafctl provider test llamacpp</code>	Confirms the HTTP provider is reachable and returns model text.
<code>nvidia-smi</code>	Confirms the NVIDIA driver and live GPU activity.

Common Failure

Observed:

```
vulkaninfo sees NVIDIA GeForce RTX 4070
llama-server --list-devices prints only:
  Available devices:
```

Meaning:

The selected llama.cpp package is CPU-only.

Fix:

```
Use a Windows x64 Vulkan llama.cpp build.
Pass its directory with --llama-dir or set LEAF_LLAMACPP_DIR.
```

Runtime Doctrine

```
The GPU proposes.
The CPU disposes.
```

```
Model output is untrusted text.
The parser turns it into a typed artifact.
The patch gate decides whether it can touch files.
The journal records what happened.
```

Promotion Gate

The provider spine is ready for 0.4.0-alpha.1 evidence when all of these are true:

1. `serve-llamacpp.ps1 -check-only -backend vulkan` sees at least one device.
2. `llama-server` stays alive as a persistent HTTP endpoint.
3. `leafctl provider test llamacpp` passes all checks.
4. Brain and coder lanes can route to `llamacpp` independently.
5. A generated patch is validated before any apply operation.